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PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light Spectrum for Long Duration Space Flight		
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Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
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COI Name (Institution):	Smith, Peter (University of Arizona) Hanifin, John (Thomas Jefferson University) Pineda, Maria (Thomas Jefferson University) Lockley, Steven (Brigham and Women's Hospital)		
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Task Description:

The goal of this project is to determine the optimal wavelengths of light for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits are a risk factor during space missions. The resulting physiological and behavioral changes from sleep and circadian disruption can threaten the success of a mission by diminishing alertness, cognitive ability, and psychomotor performance. Currently, over 45% of all medications taken in space are sleep aids. Ambient light is the primary stimulus for human circadian regulation. While on Earth, humans have a 24-hour day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either a rapidly changing or severely disrupted day/night cycle, and must work in a dimly lit spacecraft interior with few or no windows. Bright white light has already been implemented as a pre-launch countermeasure, but has yet to be used during space flight. Providing sufficient light intensities to work areas in the International Space Station (ISS), the planned Crew Exploration Vehicle (CEV), and future lunar habitats raises several concerns, among which are heat production, energy consumption, and upmass. Mitigating these factors requires a better understanding of how different wavelengths of light regulate the human circadian system. From this data it may be possible to optimize the light spectrum both as a countermeasure for sleep and circadian disturbances before or during space exploration. Listed below are the specific aims of this project. Given changes in NASA priorities and a budget cut to this project, the third specific aim was re-aligned mid-project.

1) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Our published action spectrum indicates that the blue monochromatic wavelengths (446-477 nm) are the most potent for melatonin suppression. This aim was intended to determine if increasing the power in the blue region of a polychromatic fluorescent light can increase potency for melatonin suppression while keeping the visual experience white. Philips Lighting B.V., an NSBRI industry partner, donated the prototype blue-enriched lamps and exposure systems used in this study. Since the start of the project, 25 human subjects have completed a total of 248 nighttime melatonin suppression trials. The data are now being analyzed for publication.

2) Test the hypothesis that the white polychromatic fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption have increased efficacy for circadian phase-shifting when they are enriched in the blue part of the spectrum. For this aim, we are comparing the magnitude of circadian phase-shifts in healthy subjects following exposure to equal photon densities of blue-enriched fluorescent lights versus the white lamps currently used by NASA. Each subject goes through a seven day, 24 hour a day, inpatient trial with assessment of multiple biological and behavioral endpoints. A total of 21 subjects have been entered into the protocol and of these, 20 completed the full study. Assay of blood samples, assessment of behavioral data is ongoing for this study and enrollment has not been closed.

3) Test the hypothesis that three ambient extraterrestrial light environments (skylight during low earth orbit for ISS and the CEV, ambient light on the lunar surface, and ambient light on the surface of Mars) differ in their capacity to regulate melatonin. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial and passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of non-terrestrial light exposures. To test this hypothesis, we are working collaboratively with investigators from the Lunar and Planetary Laboratory (LPL) of the University of Arizona who are experts in characterizing the light in a variety of extraterrestrial environments. Although progress was delayed by almost two years due to unforeseen optical design problems, a functional optical array was developed to establish melatonin suppression responses with polychromatic light that simulates light in low Earth orbit, ambient light on the lunar surface and ambient light on the surface of Mars. The first prototype of this optical system was delivered to our lab in March 2007. Since then, 9 human subjects have completed over 80 nighttime melatonin suppression trials with simulated lunar light and 9 other subjects have completed over 60 nighttime trials with simulated martian light. Running of subjects, assay of blood samples and data analysis is ongoing.

Although our work is transitioning to a new four years of specific aims (NRA NNJ07ZSA002N), some work on elements from the previous four years will be continued. Specifically we plan to 1) submit a manuscript for peer-review on all of the data from Aim 1, complete the phase shift study in Aim 2, and complete the fluence-response curves with simulated lunar light and mars light in Aim 3. For the new aims we plan to develop new solid state and extraterrestrial light exposure systems, develop melatonin suppression fluence-response curves with solid state light, initiate a melatonin suppression fluence-response curve with simulated Earth daylight, and start recruitment for a phase shift/alertness study with solid state light.

Ultimately, these data will impact NASA lighting requirements and improve light treatment used as a countermeasure for circadian and sleep-wake disruption in space exploration. The data will also help NASA identify the best spectral transmission for space suit visors, space vehicles, and habitat windows, and the ideal spectral distribution for illumination of living quarters.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The knowledge we hope to gain from this research, though focused on spaceflight, will also benefit people on Earth. The sleep deficits experienced by astronauts during space flight can be considered a threat to the success of space missions (Longnecker and Molins, 2005). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability and psychomotor performance (Dijk et al., 2001). As a measure to counteract sleep disruptions, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (US Bureau of Labor Statistics, 2007). Shift workers have been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, and cognitive and emotional problems. Furthermore, epidemiological studies of female shift workers have shown that they are statistically more likely to suffer from breast and colon cancer as compared to their day shift counterparts. The World Health Organization has identified shift work as a probable risk for cancer (The International Agency for Research on Cancer, 2007). Our laboratory is involved in testing the hypothesis that night time exposure to light suppresses melatonin and contributes to cancer risk (Blask et al., 2005; Stevens et al., 2007). Aside from evidence of a breakdown in physical health, the effects of circadian disruption and sleep loss have long been known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of psychomotor coordination and decreased physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers

	affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military, and more recently, homeland security. Many people also experience the same effects after air travel across several time zones. In the past, the U.S. Air Force has supported our laboratory to study the acute alerting effects of light (French et al., 1990; Brainard et al., 1996). Our current work for NIH has continued this effort (Lockley et al., 2006). Existing light therapy interventions stand to benefit from enhancing our understanding of how different wavelengths of the spectrum affect human circadian and neurobehavioral regulation. The result may be a more efficient intervention with increased potency and/or fewer side effects. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD) (Lam and Levitt, 1999). Similar bright white light interventions are also used to treat jetlag. Side effects from exposure to bright white light for these and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders may deliver the same medical impact with lower levels of light intensity, and potentially fewer side effects. Our group has completed Phase I testing of light therapy with blue solid-state lighting for SAD patients (Glickman et al., 2006).
Task Progress:	At the close of this four year funding cycle, the majority of work proposed in each aim has been completed. Aim 1: Test the hypothesis that the white fluorescent lamps currently used by NASA as a pre-launch countermeasure for circadian and sleep disruption will have increased efficacy for melatonin suppression when they are enriched in the blue portion of the spectrum. Philips Lighting B.V., an NSBRI industry partner, donated the prototype blue-enriched lamps and exposure systems used in this study. Since the start of the project, 25 human subjects have completed a total of 248 nighttime melatonin suppression trials. Three full fluence response curves have been completed. The data are now being analyzed for publication. Ultimately, the results from healthy men and women will determine if white-appearing, blue enriched fluorescent light (17, 000 Kelvin) is significantly stronger for melatonin suppression than the white-appearing fluorescent light (4,000 Kelvin) currently used by NASA as a pre-launch countermeasure for astronauts. Aim 2: Test the hypothesis that the white fluorescent lamps currently used by NASA have increased efficacy for circadian phase-shifting when they are enriched in the blue portion of the spectrum. For this aim, we are comparing the magnitude of circadian phase-shifts in healthy subjects following exposure to blue-enriched fluorescent lights versus equal photon densities of the white lamps currently used by NASA. Each subject goes through a seven day, inpatient trial with assessment of multiple biological and behavioral endpoints. To date, over 300 subjects were screened, with 21 meeting all study requirements and enrolling in the study. As proposed, 20 healthy men and women completed the full study. Assay of blood samples and assessment of behavioral data are ongoing for this study and enrollment has not been closed. The results of the study should be evident within the coming year. Aim 3: Test the hypothesis that ambient extraterrestrial light environments (low Earth orbit skylight, light on the lunar surface, and light on the surface of Mars) differ in their capacity to regulate melatonin. Since circadian and sleep regulation depends on an astronaut's total exposure to both artificial light and passive ambient illumination through windows or space suit visors, it is essential to determine the circadian effects of non-terrestrial light exposures. To test this hypothesis, we collaborated with LPL optical engineers at the University of Arizona. Although progress was delayed by almost two years due to design and fabrication problems, the first functional optical array was delivered to our lab in March 2007. To date, 9 human subjects have completed over 80 nighttime melatonin suppression trials with simulated lunar light and 9 other subjects have completed over 60 nighttime trials with simulated martian light. Running of subjects, assay of blood samples and data analysis are ongoing.
Bibliography Type:	Description: (Last Updated: 01/19/2026)
Abstracts for Journals and Proceedings	Gooley JJ, Brainard GC, Rajaratnam SM, Kronauer RE, Czeisler CA, Lockley SW. "Spectral sensitivity of the human circadian timing system." 5th World Congress of the World Federation of Sleep Research and Sleep Medicine Societies, Cairns, Australia, September 2-6, 2007. World Sleep. 2007. , Sep-2007
Abstracts for Journals and Proceedings	Gooley JJ, Rajaratnam SM, Brainard GC, Kronauer RE, Czeisler CA, Lockley SW. "Spectral sensitivity of the human circadian timing system." Aerospace Medical Association 79th Annual Meeting, Boston, MA, May 11-15, 2008. Aviation, Space, and Environmental Medicine, 2008 Mar;79(3):266-7. , Mar-2008
Abstracts for Journals and Proceedings	Brainard GC. "Phototransduction for melatonin regulation: The magic of illumination." FASEB Research Conference, Snowmass Village, CO, August 10-15, 2008. FASEB Research Conference, Abstract Book, August 2008. , Jan-2010
Abstracts for Journals and Proceedings	Gooley JJ, Rajaratnam SM, Brainard GC, Kronauer RE, Czeisler CA, Lockley SW. "The spectral sensitivity of human circadian photoreception is dynamic and changes depending on the irradiance and duration of light." Society for Research on Biological Rhythms 20th Anniversary Meeting, Destin, Florida, May 17-21, 2008. Society for Research on Biological Rhythms 20th Anniversary Meeting. Abstract Book, May 2008. , May-2008
Abstracts for Journals and Proceedings	Brainard GC, Hanifin JP, Lockley SW, Warfield B, James M, Cecil K, West K, Jablonski M, Byrne B, Rollag M. "Developing light as a countermeasure for circadian and sleep problems in space exploration." Light and Health Research Foundation Symposium on Light, Performance and Quality of Life, Eindhoven, Netherlands, November 8, 2007. Nov-07 Performance and Quality of Life, Abstract Book, November 2007. , Nov-2007
Articles in Peer-reviewed Journals	Brainard GC, Sliney D, Hanifin JP, Glickman G, Byrne B, Greeson JM, Jasser S, Gerner E, Rollag MD. "Sensitivity of the human circadian system to short-wavelength (420-nm) light." J Biol Rhythms. 2008 Oct;23(5):379-86. PMID: 18838601 , Oct-2008
Articles in Peer-reviewed Journals	Zaidi FH, Hull JT, Peirson SN, Wulff K, Aeschbach D, Gooley JJ, Brainard GC, Gregory-Evans K, Rizzo JF 3rd, Czeisler CA, Foster RG, Moseley MJ, Lockley SW. "Short-wavelength light sensitivity of circadian, pupillary, and visual awareness in humans lacking an outer retina." Curr Biol. 2007 Dec 18;17(24):2122-8. PMID: 18082405 , Dec-2007

Awards	Brainard GC. "Invited Keynote Speaker at the Light and Health Symposium, San Francisco, CA , March 13, 2008." Mar-2008
Awards	Brainard GC. "Invited Keynote Speaker at the National Light Congress, Eindhoven, Netherlands, November 8, 2007." Nov-2007
Awards	Brainard GC. "Session Chair, Gordon Research Conference on Pineal Cell Biology, Barga, Italy, April 23, 2008." Apr-2008
Awards	Brainard GC. "Thomas Jefferson University 2007 Research Performance Award, January 2008." Jan-2008
Awards	Smith P. "Leader of the current Phoenix Mars Mission, launched August 3, 2007, landed May 25, 2008, and provided first ground verification of water on Mars. This is the first Mars mission run from a university." May-2008
Books/Book Chapters	Doghramji K, Balaicuis JM, Brainard GC. "Sleep and sleep disorders." in "Integrative psychiatry." Ed. D.A. Monti, B. Beitman. New York : Oxford University Press, 2010., Jan-2010
Papers from Meeting Proceedings	James M, Brainard GC. "Photoreception for light and human health." The 26th Session of the CIE, Beijing, China, July 4-11, 2007. Proceedings of the 26th session of the CIE, p. 486-494, 2007. , Jul-2007
Papers from Meeting Proceedings	Brainard GC, Veitch JA. "Lighting and health workshop - final report." The 26th session of the CIE, Beijing, China, July 4-11, 2007. Proceedings of the 26th session of the CIE, p. 550-553, 2007. , Jul-2007
Papers from Meeting Proceedings	Jablonski MR, Hanifin JP, Rollag MD, Brainard GC. "Neuroendocrine and neurobehavioral effects of blue-enriched lighting." SOLG Light and Health Symposium, Eindhoven, Netherlands, November 8, 2007. In: Light, Performance and Quality of Life. SOLG Light and Health Symposium, p. 18-25, 2008. , Aug-2008
Significant Media Coverage	Donoff E. "Light's impact on health is playing a central role in design. This article specifically mentions NSBRI." Architectural Lighting Magazine, March 2008., Mar-2008
Significant Media Coverage	Svensson P. "This article was featured on many websites including the New York Times, CBS news, MSNBC and over 35 other notable sources. NSBRI is specifically mentioned." Associated Press, July 2008., Jul-2008